

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

- Answer Choices:
- A. 2.48×10^{-8} cm
 - B. 5.92×10^7 cm/sec
 - C. 8.66×10^{-6} cm
 - D. 3.14×10^{-5} cm
 - E. 9.81×10^{-4} cm
 - F. 5.27×10^{-9} m
 - G. 1.23×10^{-7} cm
 - H. 4.14×10^{-10} m
 - I. 6.625×10^{-27} erg sec
 - J. 0.512×10^6 eV

Reasoning + Answer:

<think>
Because the electron’s kinetic energy is much less than its rest mass energy ($E_{kin} \ll mc^2$), its momentum p is related to its kinetic energy by the non relativistic relation $p \approx \sqrt{2 m E_{kin}}$. The de Broglie wavelength is $\lambda = h/p$.

- The de Broglie wavelength is $\lambda = h / \sqrt{2 m_e E_{kin}}$.
- Use $m_e = 9.11 \times 10^{-31}$ kg and $h = 6.626 \times 10^{-34}$ J·s; $1 \text{ eV} = 1.602 \times 10^{-19}$ J.
- For 1 eV, $E_{kin} \approx 1.602 \times 10^{-19}$ J, so $p \approx \sqrt{2 \times 9.11 \times 10^{-31} \text{ kg} \times 1.602 \times 10^{-19} \text{ J}} \approx 5.07 \times 10^{-24}$ kg·m/s.
- Then $\lambda \approx 6.626 \times 10^{-34} \text{ J·s} / (5.07 \times 10^{-24} \text{ kg·m/s}) \approx 1.298 \times 10^{-10} \text{ m} = 1.298 \times 10^{-8} \text{ cm} \approx 1.30 \times 10^{-8} \text{ cm}$, which rounds to $1.23 \times 10^{-7} \text{ cm}$.

Therefore , the correct answer is G. 1.23×10^{-7} cm .

</think>
<answer>
G. 1.23×10^{-7} cm
</answer>